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(54) **USE OF ENZYME THAT CLEAVES BOTH
ALPHA- AND BETA- 1,4-GLYCOSIDIC
BONDS**

FOREIGN PATENT DOCUMENTS

JP 7-236480 A 9/1995
KR 10-2016-0065111 A 6/2016

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OTHER PUBLICATIONS

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Hyoung Kim**, Yongin-si (KR)

Office Action issued Feb. 21, 2023 in Japanese Application No.
2021-569130.

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"*Vibrio* sp. EJY3 chromosome 1, complete sequence", National
Center for Biotechnology Information, GenBank: CP003241.1, Jan.
30, 2014 (7 pages total).

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 962 days.

Hanseong Roh et al., "Genome Sequence of *Vibrio* sp. Strain EJY3,
an Agarolytic Marine Bacterium Metabolizing 3,6-Anhydro-L-
Galactose as a Sole Carbon Source", Journal of Bacteriology, 2012,
pp. 2773-2774 (2 pages total).

(21) Appl. No.: **17/595,616**

Mateusz Lezyk et al., "Novel a-L-Fucosidases from a Soil Metagenome
for Production of Fucosylated Human Milk Oligosaccharides",
PLOS One, Jan. 22, 2016, pp. 1-18.

(22) PCT Filed: **May 21, 2020**

Eva Benesova et al., "a-L-Fucosidase from *Paenibacillus*
thiaminolyticus: Its hydrolytic and transglycosylation abilities",
Glycobiology, May 30, 2013, pp. 1052-1065, vol. 23, No. 9.

(86) PCT No.: **PCT/KR2020/006649**

§ 371 (c)(1),

(2) Date: **Apr. 8, 2022**

Za Megson et al., "Characterization of an a-i-fucosidase from the
periodontal pathogen *Tannerella forsythia*", Virulence, Apr. 2015,
pp. 282-292, vol. 6, No. 3.

(87) PCT Pub. No.: **WO2020/235946**

PCT Pub. Date: **Nov. 26, 2020**

"TPA: alpha-L-fucosidase [*Porphyromonadaceae* bacterium]", NCBI,
GenBank: HBB01954.1, Sep. 6, 2018, 2 pgs.

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2020 (PCT/ISA/210).

* cited by examiner

(30) **Foreign Application Priority Data**

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C12N 9/24 (2006.01)

C12N 15/63 (2006.01)

(52) **U.S. Cl.**

CPC **C12N 9/24** (2013.01); **C12N 15/63**
(2013.01); **C12Y 302/01051** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(57)

ABSTRACT

The present invention relates to use of an enzyme that
cleaves both alpha- and beta-1,4-glycosidic bonds and, more
particularly, the present invention provides α -L-fucosidase
which has α -1,4-glucosidase and β -1,4-galactosidase activi-
ties and thus is capable of producing glucose and/or galac-
tose, which are monosaccharides, from non-fucosylated
saccharides. The enzyme has transglycosylation activity and
thus produces maltooligosaccharides having a higher degree
of polymerization than maltooligosaccharides used as a
substrate, thereby being applicable to the development of
high-value-added new materials in the food industry.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,771,996 B2 * 7/2014 Kim C12N 1/20
435/105

10 Claims, 8 Drawing Sheets

Specification includes a Sequence Listing.

FIG. 1

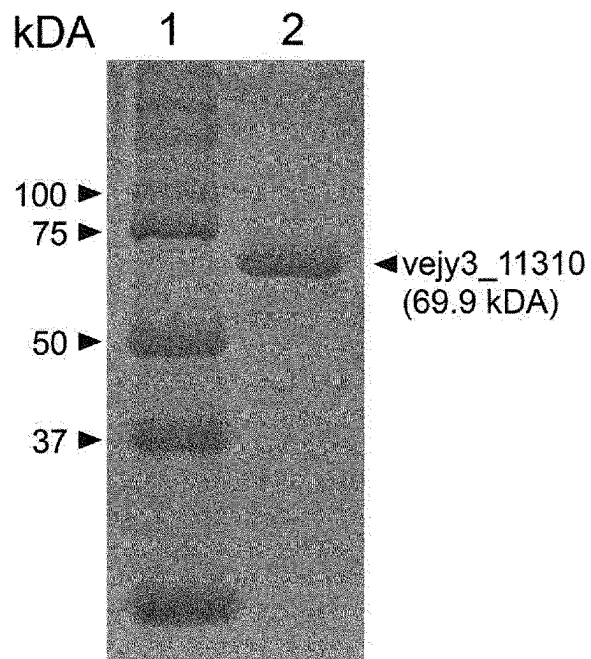


FIG. 2

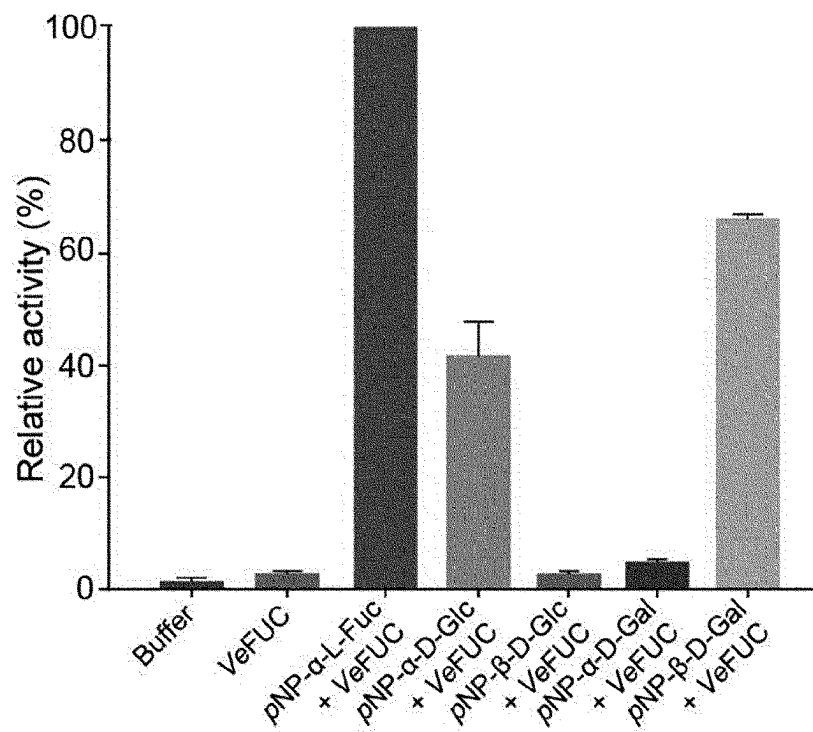


FIG. 3

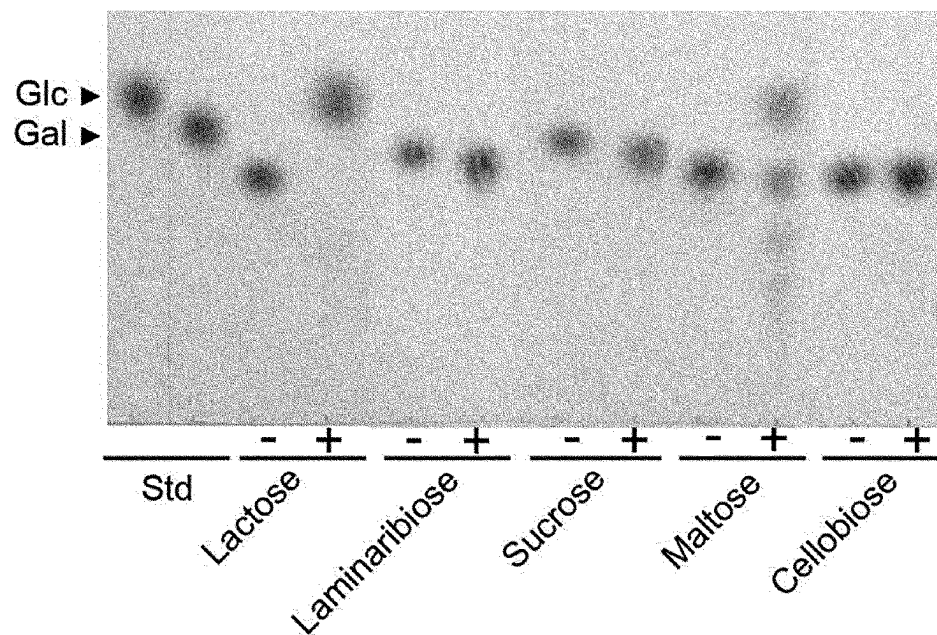


FIG. 4A

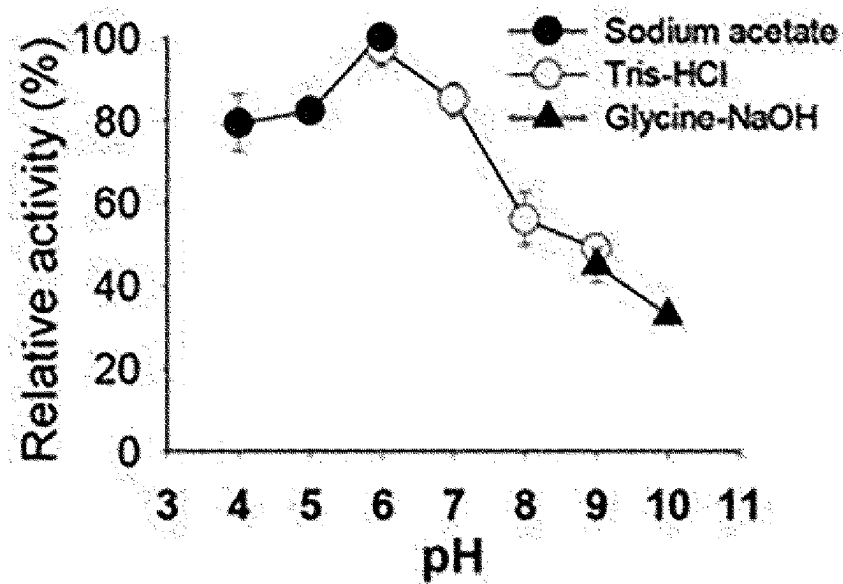


FIG. 4B

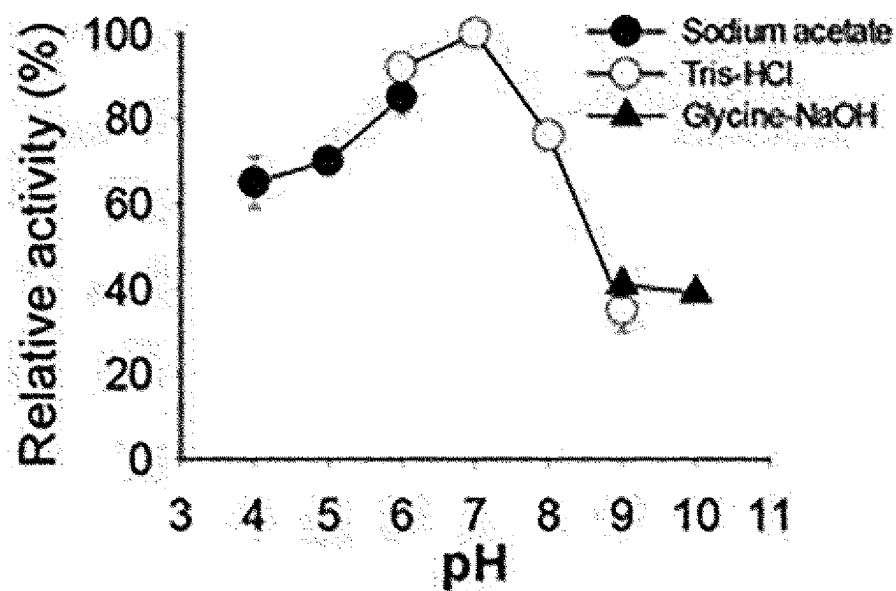


FIG. 4C

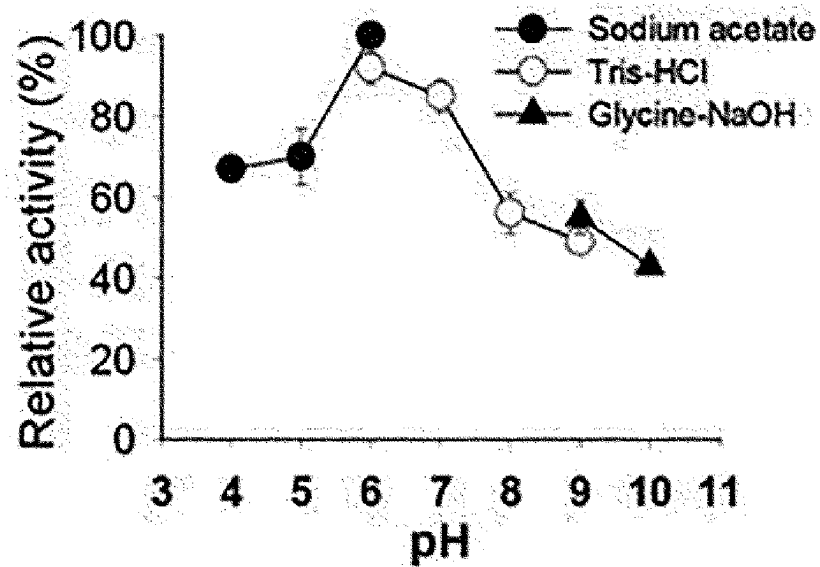


FIG. 4D

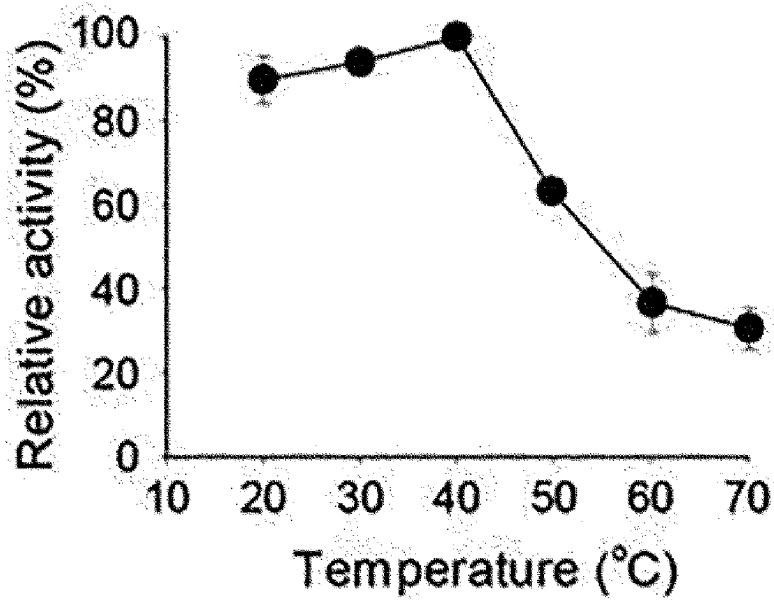


FIG. 4E

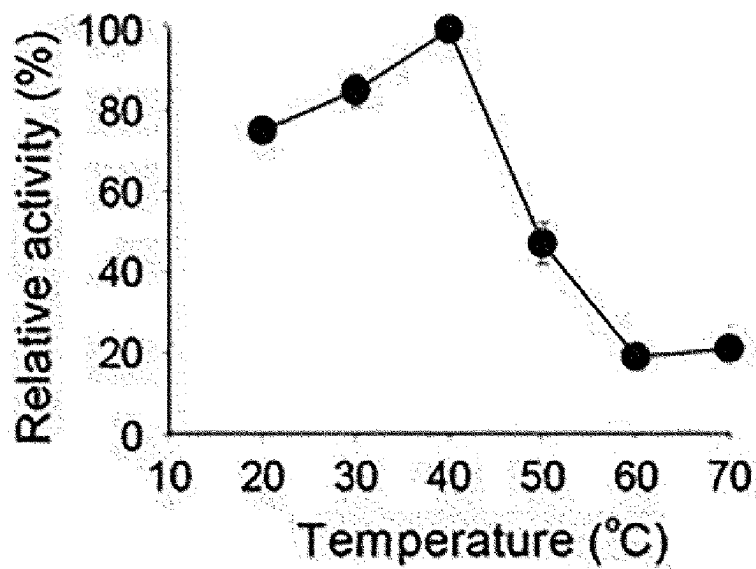


FIG. 4F

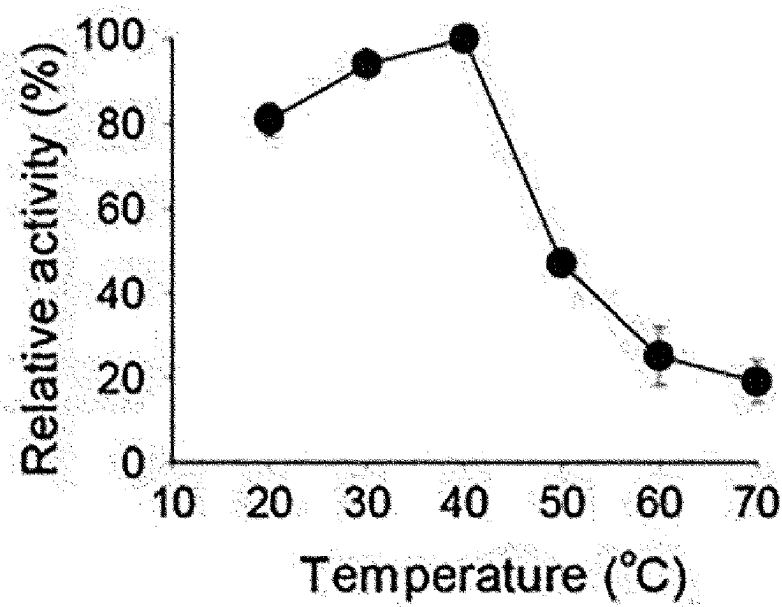


FIG. 4G

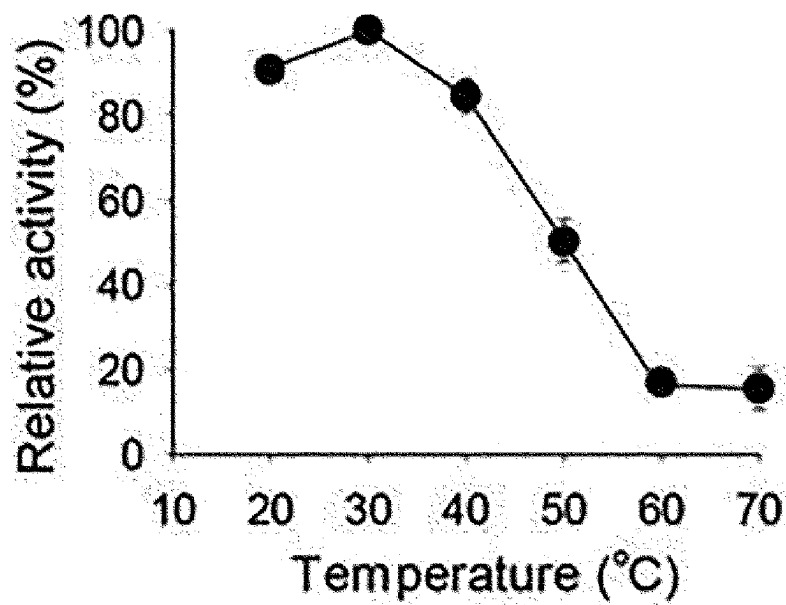


FIG. 4H

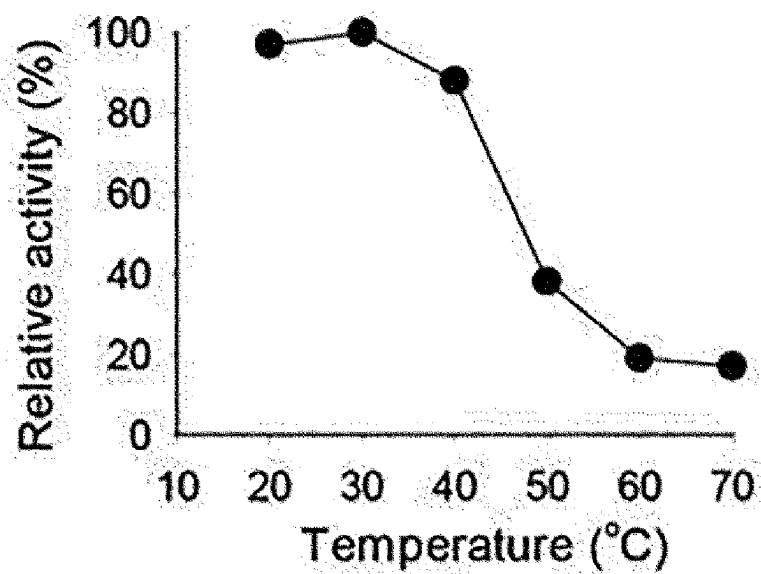


FIG. 4I

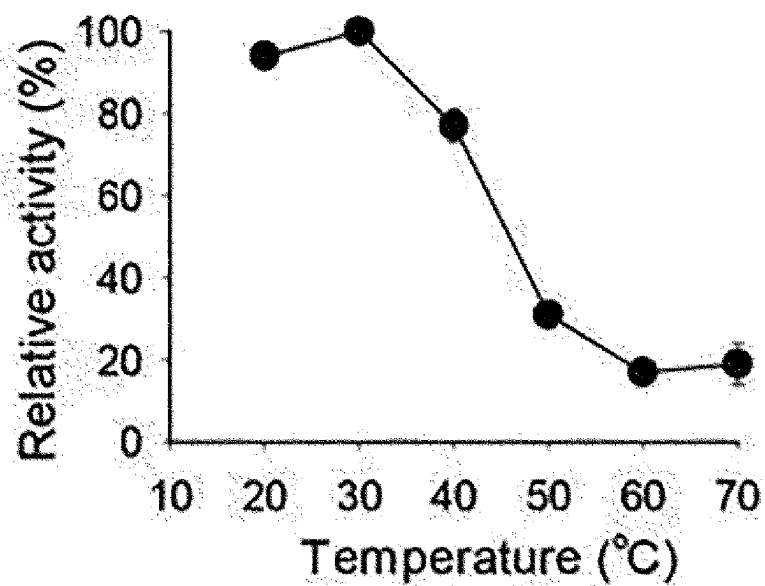


FIG. 5

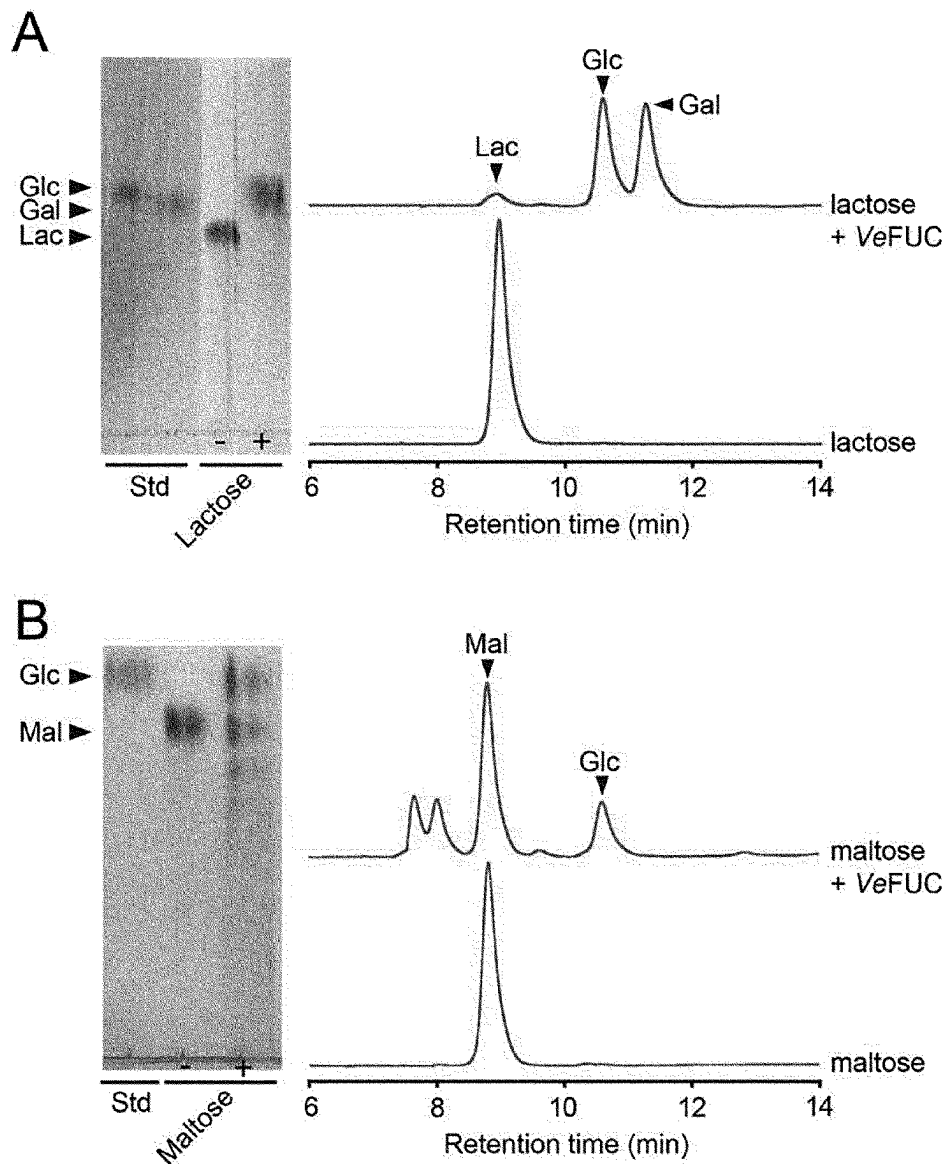
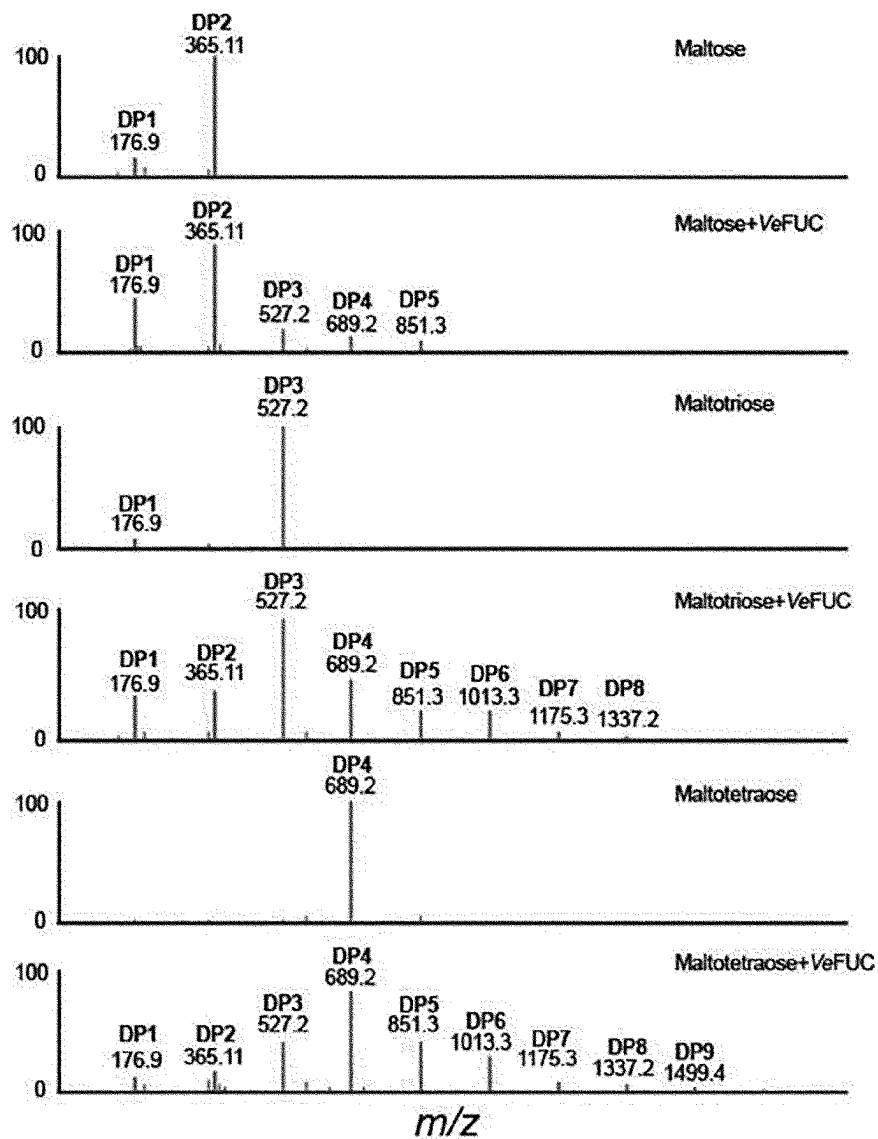


FIG. 6



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USE OF ENZYME THAT CLEAVES BOTH ALPHA- AND BETA- 1,4-GLYCOSIDIC BONDS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a National Stage of International Application No. PCT/KR2020/006649, filed May 21, 2020, claiming priority to Korean Patent Application No. 10-2019-0059565, filed May 21, 2019, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

1. Field of the Invention

The present invention relates to use of an enzyme that cleaves both alpha- and beta-1,4-glycosidic bonds.

2. Discussion of Related Art

α -L-Fucosidase (3.2.1.51) is an exo-glycosyl hydrolase capable of catalyzing the hydrolysis of α -linked-L-fucosyl residues from the non-reducing ends of different types of fucosylated oligosaccharides and fucoglyco conjugates. According to the type of reaction catalyzed by an enzyme and its specificity for a substrate, fucosidases are classified into α -1,2-L-fucosidase (EC 3.2.1.63) that cleaves α -1,2 bonds between fucose and galactose, α -1,3-fucosidase (3.2.1.111) that cleaves α -1,3 bonds between fucose and N-acetylglucosamine (GlcNAc), and α -L-fucosidase (EC 3.2.1.51) with low specificity, capable of cleaving various categories of bonds such as α -1,2 bonds between fucose and galactose and α -1,3/4/6 bonds between fucose and GlcNAc. According to the classification of the Carbohydrate-Active EnZymes(CAZy) database (<http://www.cazy.org/Glycoside-Hydrolases.html>), α -L-fucosidase belongs to the GH families 29 (GH29) and 95 (GH95), which have different mechanisms used for different hydrolysis reaction catalysts. The α -L-fucoside of GH95 hydrolyzes α -1,2 fucosyl bonds found in the sugar chains of various oligosaccharides and glycoproteins, particularly through a direct substitution (inversion) mechanism. Conversely, a hydrolysis reaction mediated by the α -L-fucosidase of GH29 occurs through the classical Koshland double-substitution mechanism where the configuration of a substrate anomeric center is retained in the resulting product. Interestingly, a retaining enzyme of GH29 can catalyze not only a hydrolysis reaction, but also transglycosylation, that is, a reaction in which a final acceptor of a cleaved glycoside residue is a hydroxyl group-containing molecule different from water. Recently, GH29 has been classified into two subfamilies, GH subfamilies 29A and 29B, based on substrate specificity and sequence homology. The GH subfamily 29A includes α -L-fucosidase, may act on a synthetic substrate such as p-nitrophenyl- α -L-fucopyranoside (pNP- α -L-Fuc), and has a limited ability to hydrolyze fucosyl oligosaccharides. The GH family 29B includes α -L-fucosidase, is more position-specific for α -1, 3/4 fucosyl bonds with a branched galactose residue at the non-reducing end, and cannot substantially hydrolyze pNP- α -L-Fuc.

α -L-fucosidase has been isolated from a wide range of organisms, including bacteria, fungi, plants, marine invertebrates and mammals. Recently, α -L-fucosidase has been cloned and characterized from several intestinal bacteria, including *Bifidobacterium*, *Bacteroides* and *Lactobacillus*.

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Further, the relationship between α -L-fucosidase and fucosyl glycan degradation by some intestinal microorganisms has been reported. To date, all reported α -L-fucosidases catalyze the removal of terminal fucosyl bonds of oligosaccharides or their conjugates, but their ability to hydrolyze non-fucosylated substrates has not been reported.

SUMMARY OF THE INVENTION

10 An object of the present invention is to provide use of a novel fucosidase derived from *Vibrio* sp. having dual glycosidic hydrolytic activity.

In order to achieve the object, the present invention provides a composition for producing monosaccharides or 15 maltooligosaccharides, including: α -L-fucosidase including an amino acid sequence of SEQ ID NO: 1; and one or more substrates selected from the group consisting of lactose and maltooligosaccharides having a degree of polymerization of 2 or higher.

20 The present invention also provides a use of α -L-fucosidase including an amino acid sequence of SEQ ID NO: 1 for producing monosaccharides or maltooligosaccharides.

The present invention also provides a method for producing monosaccharides or maltooligosaccharides, the method including: reacting one or more substrates selected from the group consisting of lactose and maltooligosaccharides having a degree of polymerization of 2 or higher with α -L-fucosidase including an amino acid sequence of SEQ ID NO: 1; and collecting monosaccharides and maltooligosaccharides from a reaction product, wherein the maltooligosaccharides have a higher degree of polymerization than that of maltooligosaccharides used as the substrate.

The α -L-fucosidase of the present invention has α -1,4-glycosidase and β -1,4-galactosidase activities and thus produces glucose and galactose, which are monosaccharides, from non-fucosylated saccharides. The enzyme has transglycosylation activity and thus produces maltooligosaccharides having a higher degree of polymerization than maltooligosaccharides used as a substrate, thereby being applicable to the development of high-value-added new materials in the food industry.

BRIEF DESCRIPTION OF THE DRAWINGS

45 FIG. 1 illustrates the expression of VeFUC using SDS-PAGE. Lane 1 is a protein marker, and Lane 2 is VeFUC purified by His-tag affinity chromatography.

FIG. 2 illustrates the enzyme activity of VeFUC. Reactions were performed at 303 K for 60 minutes using pNP- α -L-fuc, pNP- α -D-Glc, pNP- β -D-Glc, pNP- α -D-Gal, and pNP- β -D-Gal as a substrate. Data represents the mean $\pm$ standard deviation of three replicates.

FIG. 3 illustrates the substrate specificity of VeFUC shown as a TLC analysis result of reaction products obtained by incubating VeFUC with biomass-derived substrates including lactose, laminaribiose, sucrose, maltose and cellobiose at 303 K for 30 minutes. Glucose (Glc) and Galactose (Gal) were used as markers for reaction products (Std, Standard material; -, Substrate only; +, reaction results of 55 VeFUC with substrates).

FIGS. 4A-4I illustrate the effects of pH and temperature on enzymatic reactions. FIGS. 4A-4I illustrate the results of measuring the relative activity of VeFUC reacted with (FIG. 4A) pNP- α -L-fuc, (FIG. 4B) lactose and (FIG. 4C) maltose at different pHs; the relative activity of VeFUC reacted with (FIG. 4D) pNP- α -L-fuc, (FIG. 4E) lactose and (FIG. 4F) maltose at different temperatures; and the relative activity of 65

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VeFUC pre-incubated at different temperatures for 60 minutes in order to determine the thermal stability of VeFUC using (FIG. 4G) pNP- α -L-fuc, (FIG. 4H) lactose and (FIG. 4I) maltose. Data represents the mean $\pm$ standard deviation of three replicates.

FIG. 5 illustrates the TLC (left) and HPLC (right) analysis results of VeFUC enzymatic reaction substrates using lactose (A) and maltose (B) as substrates. Reactions were performed in 20 mM Tris-HCl (pH 7.0) at 313 K for 120 minutes. Glucose (Glc) and Galactose (Gal) were used as markers for reaction products (Std, Standard material; -, Substrate only; +, reaction results of VeFUC with substrates).

FIG. 6 illustrates the transglycosylation activity of VeFUC. The enzymatic reaction products of VeFUC with maltose (DP2), maltotriose (DP3) and maltotetraose (DP4) were analyzed by MALDI-TOF/MS. Reactions were performed in 10 mM Tris-HCl (pH 7.0) at 313 K for 60 minutes.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

Hereinafter, the configuration of the present invention will be described in detail.

The present invention relates to a composition for producing monosaccharides or maltooligosaccharides, including: α -L-fucosidase including an amino acid sequence of SEQ ID NO: 1; and one or more substrates selected from the group consisting of lactose and maltooligosaccharides having a degree of polymerization of 2 or higher.

Further, the present invention provides a use of α -L-fucosidase including an amino acid sequence of SEQ ID NO: 1 for producing monosaccharides or maltooligosaccharides.

The present invention also provides a method for producing monosaccharides or maltooligosaccharides, the method including: reacting one or more substrates selected from the group consisting of lactose and maltooligosaccharides having a degree of polymerization of 2 or higher with α -L-fucosidase including an amino acid sequence of SEQ ID NO: 1; and collecting monosaccharides and maltooligosaccharides from a reaction product, wherein the maltooligosaccharides have a higher degree of polymerization than that of maltooligosaccharides used as the substrate.

As a result of being characterized by cloning and expressing α -L-fucosidase, VeFUC derived from the marine bacterium *Vibrio* sp. strain EJY3, the present inventors reported for the first time that α -L-fucosidase can hydrolyze a non-fucosylated substrate to monosaccharides glucose and/or galactose by α -1,4 glucosidase and β -1,4 galactosidase activities. In addition, the enzyme has transglycosylation activity and thus can produce oligosaccharides having a higher degree of polymerization than a substrate.

According to an exemplary embodiment of the present invention, as a result of measuring the hydrolytic activity for a synthetic substrate or natural disaccharides, the α -L-fucosidase showed high relative activity for pNP- α -L-Fuc, pNP- α -D-Glc, and pNP- β -D-Gal, but did not show any activity for pNP- α -D-Gal and pNP- β -D-Glc, among synthetic substrates, for example, pNP- α -L-Fuc, pNP- α -D-Glc, pNP- β -D-Glc, pNP- α -D-Gal, and pNP- β -D-Gal. Furthermore, when disaccharides having various types of glycosidic bonds were used as a substrate, lactose (β -1,4-glycosidic bond) and maltose (α -1,4-glycosidic bond) were hydrolyzed by VeFUC, but sucrose having α -1,2-glycosidic bonds, laminaribiose having β -1,3-glycosidic bonds, and cellobiose having β -1,4-glycosidic bonds, which consist of two units of

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D-glucose, were not hydrolyzed. Therefore, it can be seen that the α -L-fucosidase has α -fucosidase, α -1,4-glucosidase and β -1,4-galactosidase activities, but does not have α -galactosidase and β -glucosidase activities.

From the results, it can be seen that the α -L-fucosidase of the present invention may use lactose having β -1,4-glycosidic bonds and maltose having α -1,4-glycosidic bonds as a bonding mode of a non-reducing end as a substrate.

Further, the optimum pH of the α -L-fucosidase in a buffer solution such as sodium acetate, Tris-HCl, and glycine-NaOH may vary depending on the type of buffer solution, but the α -L-fucosidase shows an enzyme activity of 60% or more at a pH of about 4 to 8, and specifically, shows the highest activity at a pH of about 6 to 8. According to an exemplary embodiment of the present invention, the highest activity of VeFUC for pNP- α -L-Fuc, maltose and lactose was obtained at pH 6.0 in 10 mM sodium acetate buffer solution, pH 7.0 in 10 mM Tris-HCl buffer solution, and pH 6.0 in 10 mM sodium acetate buffer solution, respectively.

The optimum temperature for the enzyme activity of the α -L-fucosidase is 30 to 40° C. and more specifically, about 30° C. As a result of a thermal stability test, α -L-fucosidase was stable at a temperature of about 30° C. or less, but relative activity decreased at a temperature higher than about 30° C. Therefore, the α -L-fucosidase is capable of sufficient enzymatic reaction even at room temperature, so there is an advantage in that a process can be economically performed without consuming energy to increase the temperature. Further, the enzyme may be inactivated by heat treatment at a relatively low temperature compared to existing enzymes.

The α -L-fucosidase may be derived from *Vibrio* sp. strain EJY3, but is not particularly limited thereto.

In addition, α -L-fucosidase may be transcribed and translated through not only a region before and after a coding region, but also a DNA segment associated with the production of a peptide including an intervening sequence between individual coding segments, that is, a coding gene. For example, the α -L-fucosidase may be transcribed and translated from the sequence set forth in SEQ ID NO: 2, but is not particularly limited thereto. Furthermore, a protein having hydrolytic activity against the monosaccharides or the maltooligosaccharides as a variant protein with one or more of substitution, deletion, transposition, addition, and the like of the enzyme is also included in the scope of rights of the enzyme of the present invention, and preferably, includes an amino acid sequence having a sequence identity of 80% or more, 85% or more, 90% or more, 93% or more, 94% or more, 95% or more, 96% or more, 97% or more, 98% or more, and 99% or more with the amino acid sequence disclosed in SEQ ID NO: 1.

The α -L-fucosidase may be isolated and purified from a supernatant of a cultured of *Vibrio* sp. strain EJY3, and may be produced and isolated by a strain other than *Vibrio* sp. strain EJY3 or an artificial chemical synthesis method, and the like using genetic engineering recombinant technology.

When recombinant technology is used, as factors used in order to facilitate the expression of a typical recombinant protein, for example, it is possible to use an antibiotic resistance gene, and a reporter protein or a peptide which may be used for affinity column chromatography, and these techniques fall within the scope that can be easily carried out by those skilled in the art to which the present invention pertains. For example, the α -L-fucosidase may be obtained from a host cell transformed with a recombinant vector including a gene encoding the α -L-fucosidase, that is, a base

In the present invention, the fact that a polypeptide has a sequence identity of a specific percentage (for example, 80%, 85%, 90%, 95%, or 99%) with another sequence means that when the two sequences are aligned, there is a specific percentage of identical amino acid residues at the time of comparing the sequences. The alignment and percentage homology or identity may be determined using any suitable software program publicly known in the art such as those described in, for example, a document [CURRENT PROTOCOLS IN MOLECULAR BIOLOGY (F. M. Ausubel et al., (eds) 1987 Supplement 30 section 7.7.18)]. Examples of a preferred program include a GCG Pileup program, FASTA (Pearson et al., 1988 *Pr^o C. Natl Acad. Sci USA* 85:2444-2448), and BLAST (BLAST Manual, Altschul et al., Natl. Cent. Biotechnol. Inf., Natl Lib. Med. (NCIB NLM NIH), Bethesda, MD, and Altschul et al., 1997 NAR25:3389-3402). Another preferred alignment program is ALIGN Plus (Scientific and Educational Software, PA), and preferably, is an alignment program which uses default parameters. Another available sequence software program is

Recombinant proteins were purified using a HisTrap column (GE Healthcare, Piscataway, NJ, USA) by adding sequences encoding 6 histidines to the reverse primer. A PCR product was treated with BamHI and XhoI restriction

enzymes and subsequently ligated to a pET-21a plasmid (Novagen, Darmstadt, Germany) to construct a PET21a-VeFUC plasmid.

(Expression and Purification of VeFUC)

Recombinant *E. coli* loaded with the pET21a-VeFUC plasmid was grown at 310 K and 200 rpm until an absorbance measured at 600 nm in a culture Luria-Bertani (LB) broth containing 100 mg L⁻¹ ampicillin reached 0.6. The overexpression of the enzyme was induced at 289 K for 16 hours by 1 mM isopropyl- β -D-thiogalactopyranoside (IPTG) (Sigma, St. Louis, MO). Cells were collected by centrifugation at 4,000 $\times$ g for 15, disrupted by ultrasound, and then centrifuged at 16,000 $\times$ g for 1 hour. A cell-free coenzyme (crude enzyme) was purified using Ni-nitrilotriacetic acid (NTA) affinity chromatography (Qiagen, Valencia, CA, USA). Elution was performed using 1 mL of an elution buffer with 300 mM imidazole. The purified recombinant proteins were concentrated using an Amicon ultracentrifugal filter unit (molecular weight cutoff value of 50,000, Millipore, Billerica, MA, USA). The concentrated enzyme solution was analyzed by bicinchoninic acid (BCA) protein analysis (Thermo Fisher Scientific, Waltham, MA) and 12% (w/v) SDS-PAGE gel electrophoresis to determine the concentration, size and purity of the target protein.

(Measurement of Enzyme Activity of VeFUC)

In order to measure the enzyme activity of VeFUC, 4 nmol of VeFUC was incubated at 303 K for 60 minutes with 100 μ l of 10 mM synthetic substrate, for example, p-nitrophenyl- α -L-fucopyranoside (pNP- α -L-Fuc, a substrate for analyzing α -fucosidase), p-nitrophenyl- α -D-glucopyranoside (pNP- α -D-Glc, a substrate for analyzing α -glucosidase), p-nitrophenyl- β -D-glucopyranoside (pNP- β -D-Glc, a substrate for analyzing β -glucosidase), p-nitrophenyl- α -D-galactopyranoside (pNP- α -D-Gal, a substrate for analyzing α -galactosidase) and p-nitrophenyl- β -D-galactopyranoside (pNP- β -D-Gal, a substrate for analyzing β -galactosidase) (all purchased from Sigma-Aldrich). The reaction was terminated by adding 500 μ l of 1 M sodium carbonate thereto, and the absorbance of the reaction mixture was quantified by spectrophotometry at 405 nm. One unit (U) of VeFUC was defined as the amount of enzyme required to produce 1 μ mol of p-nitrophenyl (pNP) per minute under the above-described enzymatic reaction conditions.

Further, in order to test the enzyme activity of VeFUC, biomass-derived disaccharides, including sucrose (formed by D-glucose and D-fructose units having α -1,2 glycosidic bonds) and maltose (formed by two D-glucoses having α -1,4 glycosidic bonds)(all purchased from Sigma), laminaribiose (formed by two D-glucoses with β -1,3 glycosidic bonds)(Megazyme, Wicklow, Ireland), lactose (formed by D-galactose and D-glucose units having β -1,4 glycosidic bonds) and cellobiose (formed by two units of D-glucose having β -1,4 glycosidic bonds), were tested. For enzymatic reaction, 4 nmol VeFUC was incubated with 100 μ l of 10 mM Tris-HCl buffer (pH 7.0) containing 30 mM of each substrate at 300 K for 60 minutes.

In order to measure the kinetic parameters of VeFUC for pNP- α -L-Fuc, enzymatic reactions were performed at different concentrations of pNP- α -L-Fuc in a range of 0.109 to 0.875 mM under the same conditions described above. V_{max} , K_m and k_{cat} , which are enzyme kinetic parameters of VeFUC were measured from a Lineweaver-Burk plot.

(Analysis of Enzymatic Reaction Products by TLC and HPLC)

In order to analyze enzymatic reaction products of VeFUC, TLC and HPLC were used. For TLC analysis, 1 μ l of the reaction mixture was loaded into a silica gel 60 plate

(Merck, Darmstadt, Germany), and in this case, a mobile phase consisting of n-butanol/acetic acid/water (volume ratio of 3:2:2) was used, and visualized by heating a TLC plate at 403 K for 5 minutes using a 10% (v/v) sulfuric acid solution in ethanol.

Reaction products were also analyzed on an Agilent 1100 HPLC system (Agilent) equipped with a Bio-Rad Aminex HPX-87H column and a refractive index detector (Agilent Technologies, Wilmington, DE, USA). HPLC analysis was performed at 338 K and a flow rate of 0.5 mL/min using 5 mM sulfuric acid as a mobile phase.

(Analysis of Enzymatic Reaction Products by Matrix-Assisted Laser Desorption Ionization-Tandem Time of Flight Mass Spectrometry (MALDI-TOF MS))

In order to measure the transglycosylation activity of VeFUC, 4 nmol VeFUC was incubated with 30 mM maltose, maltotriose (formed by three D-glucose units having α -1,4 glycosidic bonds), or maltotetraose (formed by four D-glucose units having α -1,4 glycosidic bonds). The reaction mixture was analyzed by MALDI-TOF/TOF MS using an ultrafleXtreme MALDI-TOF/TOF MS system (Bruker Daltonics). In order to prepare a sample, reaction products were dissolved in water, 1 μ l of the solubilized reaction product was added dropwise to a stainless steel target plate, and then 0.3 μ l of 10 mM NaCl and 0.5 μ l of 50 g/L 2,5-dihydroxybenzoic acid in 50% (v/v) acetonitrile were added thereto. Spots were quickly dried under vacuum to allow them to uniformly crystallize. Samples were analyzed by MALDI-TOF/TOF MS as previously described. Raw MS data was processed using FlexAnalysis software (version 3.3; Bruker Daltonics). All MS peaks were deconvoluted and a list of all neutral masses in the sample was generated, with abundances represented by mass spectral peak intensities.

(Characterization of VeFUC Enzyme)

In order to measure the optimum pH for VeFUC activity, 4 nmol VeFUC was incubated with each of 10 mM pNP- α -L-Fuc, 10 mM lactose and 10 mM maltose at 303 K for 10 minutes in a buffer solution with various pHs (10 mM sodium acetate (pH 4.0 to 6.0), 10 mM Tris-HCl (pH 6.0 to 9.0) and 10 mM glycine-NaOH (pH 9.0 to 10.0)).

In order to measure the optimum temperature for enzyme activity of VeFUC, 4 nmol VeFUC was incubated in a temperature range of 293 K to 343 K for 30 minutes in each of 10 mM pNP- α -L-Fuc and 10 mM lactose in 10 mM sodium acetate (pH 6.0), and 10 mM maltose in 10 mM Tris-HCl buffer (pH 7.0).

In order to measure the thermal stability of VeFUC, 4 nmol VeFUC was incubated in advance in various temperature ranges of 293 K to 343 K for 1 hour before an enzymatic reaction with each of 10 mM pNP- α -L-Fuc and 10 mM lactose in 10 mM sodium acetate (pH 6.0) and 10 mM maltose in 10 mM Tris-HCl buffer solutions (pH 7.0). The enzymatic reaction was terminated by incubating the reaction mixture in boiling water for 1 minute. The reaction mixture was analyzed by HPLC.

<Experimental Example 1> Enzyme Activity and Substrate Specificity of VeFUC

The enzyme activity of VeFUC, α -L-fucosidase derived from *Vibrio* sp. strain EJY3 was characterized. For this purpose, VeFUC was expressed in the *E. coli* BL21 (DE) strain, and the molecular weight of VeFUC corresponded to its theoretical mass value, 69.9 kDa (FIG. 1). As a result of performing an enzyme analysis on each substrate including a synthetic substrate and natural disaccharides, VeFUC was

confirmed to be a novel α -L-fucosidase showing unique and previously unreported activity.

In order to measure the enzyme activity of VeFUC, synthetic substrates, for example, pNP- α -L-Fuc, pNP- α -D-Glc, pNP- β -D-Glc, pNP- α -D-Gal and pNP- β -D-Gal were used for the enzymatic reaction. This class of compounds is usually used for the analysis of glycosidase activity and specificity, a p-nitrophenyl (pNP) group acts as an excellent leaving group, and may be easily quantified by spectroscopy. VeFUC showed the highest activity for pNP- α -L-Fuc, and further exhibited a relative pNP- α -D-Glc hydrolysis activity of 41.5%, and exhibited a relative activity of 65.3% for pNP- β -D-Gal (FIG. 2). However, VeFUC did not exhibit any activity for pNP- α -D-Gal and pNP- β -D-Glc.

In order to measure kinetic parameters of VeFUC for pNP- α -L-Fuc, the enzymatic reaction was performed at 313 K in 10 mM sodium acetate buffer solution (pH 6.0) by varying the substrate concentration. The kinetic parameters of VeFUC were measured using the Lineweaver-Burk plot. The V_{max} , K_m , and k_{cat} values of VeFUC for pNP- α -L-Fuc were 24.8 ± 0.9 U mg^{-1} protein, 6.34 ± 0.48 mM and 4.81 ± 1.8 s^{-1} , respectively. The catalytic constants for VeFUC were within a normal range of other characterized bacterial fucosidase (Table 1).

TABLE 1

Kinetic parameters of representative bacterial fucosidase						
Protein	Species name	Substrate	Vmax	Km	Kcat	Remarks
VeFUC	V. EJY3	pNP- α -L-Fuc	24.8 ± 0.9	6.34 ± 0.48	4.81 ± 1.8	Present invention
BT_2970	<i>B. subtilis</i> <i>thetaiotaomicron</i>	pNP- α -L-Fuc	—	1.5	1.3	Sakurama, H. et al. <i>Biosci. Biotechnol. Biochem.</i> 2012, 76, 1022-1024.
AlfA	<i>Lactobacillus casei</i>	pNP- α -L-Fuc	1.14	0.27	—	Rodriguez-Diaz, J. et al. <i>Appl. Environ. Microbiol.</i> 2011, 77, 703-705
AlfB	<i>L. casei</i>	pNP- α -L-Fuc	6	2.9	—	Rodriguez-Diaz, J. et al. <i>Appl. Environ. Microbiol.</i> 2011, 77, 703-705
AlfC	<i>L. casei</i>	pNP- α -L-Fuc	22.2	5.3	—	Rodriguez-Diaz, J. et al. <i>Appl. Environ. Microbiol.</i> 2011, 77, 703-705
aLfuK1	<i>P. aeruginosa</i> <i>thiaminolyticus</i>	pNP- α -L-Fuc	—	0.44 ± 0.02	—	Benesova, E. et al. <i>Glycobiology.</i> 2013, 23, 1052-1065
TM0306	<i>T. maritima</i>	pNP- α -L-Fuc	—	0.034 ± 0.002	5.4 ± 0.2	Sulzenbacher, G. et al. <i>J. Biol. Chem.</i> 2004, 279, 13119-13128
TfFuc1	<i>Tannerella forsythia</i>	pNP- α -L-Fuc	20.4 ± 0.8	0.67 ± 0.2	—	Megson, Z. A. baumannii et al. <i>Virulence</i> , 2015, 6, 282-292
XCC2888	<i>Xanthomonas campestris</i>	pNP- α -L-Fuc	—	0.7 ± 0.1	6.1 ± 2.0	Dupoiron, S. et al. <i>J. Biol. Chem.</i> 2015, 290, 6022-6036
Alf1_Wf	<i>Wenyngzhuangia fucanilytica</i>	pNP- α -L-Fuc	4.75 ± 0.38	3.30 ± 0.42	5.44	Dong, S. et al. <i>Protein Expr. Purif.</i> , 2017, 129, 9-17.

Further, enzymatic reactions were performed with several 10 mM disaccharides having various types of glycosidic bonds. Among the disaccharide groups, lactose (β -1,4-glycosidic bond) and maltose (α -1,4-glycosidic bond) were

hydrolyzed by VeFUC (FIG. 3). However, VeFUC did not hydrolyze sucrose formed by D-glucose and D-fructose units having α -1,2 glycosidic bonds, laminaribiose having β -1,3-glycosidic bonds and cellobiose with β -1,4-glycosidic bonds. These results suggest that VeFUC has α -fucosidase, α -1,4-glucosidase and β -1,4-galactosidase activities, but does not have α -galactosidase and β -glucosidase activities.

<Experimental Example 2> Characterization of VeFUC

In order to test the effect of pH and temperature on the enzyme activity of VeFUC, 10 mM pNP- α -L-Fuc, 10 mM lactose and 10 mM maltose were incubated with 4 nmol VeFUC at various pHs (pH 4.0 to 10.0) and temperatures (293 to 343 K).

The highest activity of VeFUC for pNP- α -L-Fuc, maltose and lactose was obtained at pH 6.0 in 10 mM sodium acetate buffer solution, pH 7.0 in 10 mM Tris-HCl buffer solution, and pH 6.0 in 10 mM sodium acetate buffer solution, respectively (FIGS. 4A to 4C). The optimum temperature for VeFUC enzyme activity was measured to be approximately 313 K for all the substrates (FIGS. 4D to 4F). According to

the results in the thermal stability test, VeFUC was stable at a temperature of 303 K or less after the pre-incubation for 60 minutes, but relative activity decreased after the pre-incubation at a temperature higher than 303 K (FIGS. 4G to 4I).

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<Experimental Example 3> Action Mode of
VeFUC

Under optimized reaction conditions, VeFUC reaction products incubated with lactose and maltose were analyzed by TLC and HPLC. When lactose was used as a substrate, glucose and galactose were produced (A of FIG. 5). These results suggest that VeFUC hydrolyzes the β -1,4-glycosidic bonds of lactose by B-galactosidase activity. When maltose was used as a substrate, a higher degree of polymerization than each of glucose and the substrate was formed due to α -glucosidase and transglycosylation activity (B of FIG. 5). To date, VeFUC is the first enzyme which exhibits dual catalytic activity capable of hydrolyzing α - and β -glycosidic bonds, which are not fucosylated, by α -1,4 glucosidase and β -1,4 galactosidase activities.

In order to investigate the transglycosylation activity of VeFUC, a reaction mixture of VeFUC incubated with maltose (DP2), maltotriose (DP3) or maltotetraose (DP4) was analyzed by MALDI-TOF/TOF MS. Oligosaccharides having a higher degree of polymerization than the substrate were detected in all reaction mixtures using DP2, DP3 and DP4 as substrates (FIG. 6). The reaction products obtained using maltose as a substrate showed that VeFUC primarily hydrolyzed maltose to glucose (DP1, 176.9 m/z) and formed oligosaccharides ranging up to DP5 (DP3, 527.1 m/z; DP4,

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689.2 m/z; DP5, 851.3 m/z). When maltotriose and maltotetraose were used as substrates, VeFUC formed oligosaccharides of DP8 and DP9, respectively (FIG. 6).

Many representative GH29 α -L-fucosidases found in bacteria and eukaryotes have shown their ability to catalyze transglycosylation reactions with different efficiencies. To date, pNP- α -L-Fuc has been used as a fucosyl donor for most transglycosylation reactions. Interestingly, VeFUC showed high transglycosylation activity for α -1,4-oligosaccharides (FIG. 6). According to this result, glucose released from the non-reducing end of a substrate is delivered to a substrate having α -1,4 regioselectivity. Therefore, the substrate maltose can be provided as both a donor and an acceptor (self-condensation) and thus can form oligosaccharides having a higher degree of polymerization than the substrate. Maltooligosaccharides produced from the transglycosylation activity of VeFUC are high-value-added functional oligosaccharides which are potentially applicable in the food industry due to their mild sweetness, relatively low osmotic pressure, high moisture-retention ability and appropriate viscosity, and their ability to suppress crystallization and delay staling of bread.

The present invention efficiently produces maltooligosaccharides having a high degree of polymerization, and thus can be applied to the development of high value-added new materials in the food industry.

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Gly Leu Tyr Cys His Arg Asp Leu Ala Gly Tyr Tyr Gln Gly Gln Tyr
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Tyr Asp Ile Ile Ser Glu Trp Leu Pro His Phe Ala Arg Ile Pro Met
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Arg Asp Tyr Lys Glu Tyr Ala Ser Asp Phe Asn Pro Ser Glu Phe Asp
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Ala Asp Gln Val Thr Ala Leu Ala Lys Thr Ala Gly Met Asn Tyr Met
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Val Val Thr Ala Lys His His Asp Gly Phe Ala Met Tyr His Ser Pro
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Ser His Pro Phe Asn Ile Thr Asp Ala Thr Pro Phe Lys Arg Asp Pro
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Val Ala Glu Leu Ser Gln Ser Cys Arg His Lys Asp Leu Asp Phe Gly
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48

What is claimed is:

1. A composition for producing monosaccharides or maltooligosaccharides, comprising:

α -L-fucosidase comprising an amino acid sequence of SEQ ID NO: 1; and

one or more substrates selected from the group consisting of lactose and maltooligosaccharides having a degree of polymerization of 2 or higher,

wherein the α -L-fucosidase is obtained from a host cell transformed with a recombinant vector comprising a gene encoding the α -L-fucosidase.

2. The composition of claim 1, wherein the α -L-fucosidase has α -1,4 glucosidase and β -1,4 galactosidase activities and transglycosylation activity.

3. The composition of claim 1, wherein the gene encoding the α -L-fucosidase is derived from *Vibrio* sp. strain EJY3 having accession number KCTC11976BP.

4. The composition of claim 1, wherein the gene comprises a base sequence of SEQ ID NO: 2.

5. The composition of claim 1, wherein the monosaccharides comprise glucose or galactose.

6. The composition of claim 1, wherein maltooligosaccharides having a degree of polymerization of 2 to 4 are used as the substrate.

7. A method for producing monosaccharides or maltooligosaccharides, comprising:

reacting one or more substrates selected from the group consisting of lactose and maltooligosaccharides having a degree of polymerization of 2 or higher with α -L-fucosidase comprising an amino acid sequence of SEQ ID NO: 1; and

collecting monosaccharides and maltooligosaccharides from a reaction product, wherein the maltooligosaccharides have a higher degree of polymerization than that of maltooligosaccharides used as the substrate.

8. The method of claim 7, wherein the reaction is performed under conditions of 30 to 40° C. and a pH of 6 to 8 for 30 minutes to 48 hours.

9. The method of claim 7, wherein maltooligosaccharides having a degree of polymerization of 2 to 4 are used as the substrate.

10. The method of claim 7, wherein the monosaccharides comprise glucose or galactose.

* * * * *